PRACTICE TRAINING

How we learn before we build

Discovery: finding which problems are worth solving, and which Solutions actually work, before we spend delivery time on them.

Where we are, and where we want to be

WHAT IT FEELS LIKE TODAY

- A request arrives as a solution: "we need a portal."
- We build what was asked for, then learn whether it helped.
- When it doesn't help, the work is already sunk.
- Delivery dates get promised before anyone knows what to build.

WHERE WE WANT TO BE

- Start from an outcome, not a feature.
- Hear from the people the service is for before we decide.
- Try more than one idea, and test the riskiest part cheaply.
- Commit delivery time only to what has earned it.

Most software fails because it's based on wishful thinking or fear. Discovery grounds the work in what's actually true.

Where discovery fits

Step 4 of the process. It starts when quarterly planning makes an Initiative active.

Discovery	Delivery
Explore options	Commit to scope
What could we build?	What will we build?
Many solutions per opportunity	One Solution chosen
Tests to learn	Stories to build

Timeboxed

Most rounds take 4 to 8 weeks, never more than a quarter. When time runs out, decide with what you know.

Paid for from Strategic time

Discovery is real work. It comes out of the same share as building.

The product trio

Three perspectives in the room from the first session until the first slice is ready.

Product manager

Owns the Initiative, the brief and the decision. Asks: is it valuable, and does it move the goal?

UX designer and researcher

Plans the research, runs listening sessions, builds the maps. Asks: will people want it, and can they use it?

A developer or technical architect

Checks feasibility early and runs technical tests. Asks: can we build it, and what would it take?

Content strategist

Joins when the problem involves instructions, forms or wording.

Product owner

Takes the story map into slicing.

Advisory and steering

Advisory reviews what we learned. Steering sees the decision on the roadmap.

The path through a discovery round

1. **Pick the outcome**
2. **Plan the research**
3. **Listen to people**
4. **Map opportunities**
5. **Generate several solutions**
6. **Test assumptions**
7. **Decide**
8. **Move into slicing**

Steps 1 to 4 are the problem space. Steps 5 to 8 are the solution space. Most teams skip straight to 5.

STEP 1

Start from an outcome, not a feature

A goal from the roadmap, stated as a change in behavior you can measure.

A feature

"Build an online document checklist."

An outcome

"Zero multi-trip experiences for standard new driver licenses."

DMV GOAL 2

Baseline: 85% of applicants need more than one trip to get a new driver license. The Initiative: "We believe repeat trips represent an opportunity to advance Goal 2 by helping applicants know they are ready before they come in."

If the Initiative can't name the goal it advances, send it back.

STEP 2

A research plan is a list of questions

Not features. The questions we need answered to reduce risk.

DMV research questions

- What sends people home without a license?
- How do applicants decide what to bring?
- Where do staff and the website disagree?
- What happens between passing the knowledge test and the drive test?

Choose people by how they think

Pick participants by behavior and mindset, not demographics. Someone who over-prepares and someone who shows up hoping for the best need different things.

Include the people who serve them

Counter staff see every failed visit. They are a research source, not just a channel.

STEP 3

Listen, don't pitch

One-on-one listening sessions about what people actually did.

Ask about the last time

"Walk me through the last time you tried to get your license." Specific past stories beat opinions about the future.

Write down their words

Especially verbs and feelings. Those words become the opportunities, the map, and later the language in the service.

Don't show ideas

The goal is their thinking and guiding principles, not their reaction to our solution.

DMV

12 listening sessions: 5 first-time applicants and parents of teen applicants, 4 adults new to Oregon, 3 counter staff. Plus one week of counter staff logging every repeat visit.

A week of staff tally marks often tells you more than a month of opinions.

STEP 4

Sort what you heard

Goals, priorities, tasks, motivators and barriers.

Goal	What they're trying to achieve	**"Get my license in one trip."**
Priority	What matters most when things compete	**"I can't take another morning off work."**
Task	What they actually do today	**Read the website list, then gather papers from home.**
Motivator	What pulls them forward	**Independence: driving to school or a new job.**
Barrier	What gets in the way	**"I didn't know my copy wasn't good enough."**

Barriers are where opportunities hide. Name the top three.

STEP 4

Find the gaps, in their words

Map how people think about the task. Where nothing helps them today, there's an opportunity.

A mental model map

Their world, not our product. Group what people do and think into towers, then draw a line and show what supports each tower today.

An opportunity is not a feature request

It's an unmet need, pain or desire, phrased the way a person would say it.

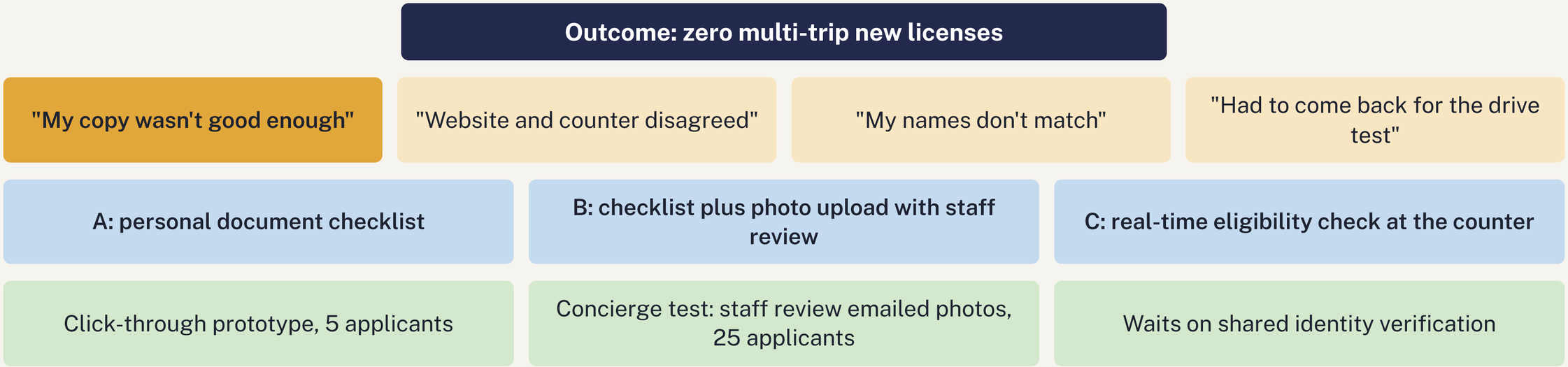
DMV TOP OPPORTUNITIES

Target	**"I didn't know my copy wasn't good enough."**
	"The website and the person at the counter told me different things."
	"My name doesn't match on two of my documents."
	"I passed the test but had to come back for the drive test."

STEP 5

Hang it on a tree

Outcome, opportunities, solutions, tests. At least three solutions for the target before you pick one.



Three solutions or you're not done. If the team can name only one, it hasn't explored the opportunity yet.

STEP 6

Test what would hurt most if it were wrong

For each serious solution: what must be true for it to work?

Desirable	Viable	Feasible	Usable
People want it	It moves the goal	We can build it	People can use it

Use the cheapest test that would change your mind

- A staff log or tally
- A paper or click-through prototype
- A concierge test: people do by hand what the software would do later
- A short technical spike

DMV: the assumption log

#	Solution	Assumption	Type	Test	Result	Status
A1	**All**	**Most repeat trips come from document problems**	Viable	One-week counter log	6 of 10 were document problems	**Holds**
A2	**B**	**Applicants will send photos of documents before a visit**	Desirable	Concierge test, 25 applicants	19 of 25 sent photos; 7 issues caught early	**Holds**
A3	**A**	**A self-serve checklist is enough on its own**	Usable	Click-through prototype, 5 applicants	3 of 5 picked the wrong birth certificate type	**Broken**
A4	**B**	**Identity verification can take pre-visit uploads**	Feasible	Two-day architect spike	Possible, needs a new intake queue	**Holds**

A3 broke, and that's a good result. It's why staff review is part of the Solution and a checklist alone was dropped.

STEP 7

Decide, and write down why

Every Solution tested gets one of three results.

Build it

B: photo upload with staff review. Document problems cause 6 of 10 repeat trips, and applicants will send photos.

Don't build it

A: the checklist alone. It failed in prototype testing.

Try something different

C waits until shared identity verification can check eligibility. The drive test gap goes to the idea log.

"Don't build" is finished work, not a failure. The Initiative carries on with its next option. No delivery date before the decision.

STEP 8

Hand off through a story map

The trio and the product owner map the chosen Solution, then slice it.

Find out what I need	Gather documents	Check documents	Book visit	Finish at the counter

First release slice

Photo upload and staff review for the two most common documents: birth certificates and proof of address.

Learning slice

Name mismatch cases, handled by hand for 20 applicants, to test the riskiest remaining assumption.

From here, the mapping and slicing deck takes over. Solution: License Ready Check, in Next from Q3 planning.

One page: the discovery brief

It lives with the Initiative. Research, maps and the tree are linked as evidence, not pasted in.

Initiative and outcome
Which goal, and the baseline

Timebox and team
Dates and who is in the trio

Research questions
What we needed to learn

Who we listened to
Sessions, logs, observations

Top opportunities
In their words; the target marked

Solutions considered
At least three

Riskiest assumptions
Linked to the assumption log

Decision and into slicing
Build, don't build, or try different, and why

Link research, don't paste it. The brief holds the decision and the evidence trail on one page.

Rules of thumb

Start from an outcome.
If the Initiative can't name its goal, send it back.

Listen before you pitch.
Ask about the last time, in their words.

Three solutions or you're not done.
One idea means the opportunity wasn't explored.

Test the riskiest assumption first.
Use the cheapest test that would change your mind.

Keep a developer in the room.
Feasibility found in week one is cheap.

Timebox it, then decide.
Discovery with no decision date turns into endless research.